Analysis of anomalies or outlier transactions of Priority Bank Mandiri customers

Data PIC	Iqbal Yusra Karim
Analysis Date	October 2, 2025
Analysis Status	Done
Requestor	Ari
Type	Deep Analysis

- Task Explenation
- Objective
- Success Metric
- Documents
- Result (Fill by Analyst)
    - Conclusion/TLDR
    - Finding/Analysis
- Recommendation (Filled by Analyst)
- Section for Data Team

Task Explanation

Bank Mandiri aims to detect anomalies or outlier transactions within the priority customer segment to enhance risk monitoring, prevent fraud, and provide personalized handling for high-value clients. Using high-value transaction data, participants are asked to identify outliers and profile the corresponding customers.

Questions:

1. Who are the customers with the highest-value transactions?
2. Which products and channels are most frequently used by priority customers?
3. In which branch do the highest-value transactions occur?
4. How can the transaction distribution be visualized and outliers detected using a box plot?
5. What is the demographic profile of customers who frequently make large transactions?

Objective:

To identify unusual (outlier) transactions and understand the characteristics of high-value customers across different regions.

Success Metric

Achieve **≥85% accuracy** in identifying high-value or anomalous transactions compared to manual verification.

Documents

https://drive.google.com/drive/folders/1neiTXSphQhNx00RxEzLQ5UKvFp5oAz49

Result

Conclusion/ TDLR:

- Q1. **Putri Simbolon** – IDR **3.39 billion, Gawati Putra** – IDR **3.00 billion** and **drg. Hasan Waluyo** – IDR **2.94 billion.** These three customers are identified as **high-value clients** who contribute significantly to the total transaction volume within the priority customer segment.
- Q2. This behavior may indicate are **higher trust in manual transactions**, especially for large sums. And **Limited adoption or comfort with digital banking** within this segment. The dominance of savings accounts also shows that many customers focus on liquidity rather than long-term investment products.
- Q3. Yogyakarta recorded the highest total transaction value among all branches, indicating a concentration of high-value clients.
- Q4. Based on the boxplot of "Transaction Amount Distribution and Outliers," the data shows a **balanced and stable distribution** with **no data points outside the whiskers**. This indicates that **no extreme or abnormal transactions** are present in this dataset. The first quartile (Q1) is approximately **IDR 125 million**, the median is **IDR 255 million**, and the third quartile (Q3) is **IDR 377 million**, suggesting that most customer transactions fall within a reasonable range.
- Q5.By Gender: Both male (L) and female (P) customers recorded nearly equal highest transaction values — approximately IDR 499.96 million and IDR 499.99 million, respectively. This suggests that gender does not significantly influence the maximum transaction capability among priority customers.
- Q5. By Age Group:The highest transaction values are nearly identical across all age categories:
  - <25: IDR 498.54 million
  - 25–35: IDR 499.91 million
  - 36–50: IDR 499.96 million
  - 51–65: IDR 499.99 million
  - 65+: similar range (slightly below).

  This shows that high-value transactions are evenly distributed across different age groups, with the 51–65 group slightly leading.

Finding/Analysis:

Q1. The customer with the tallest bar has the highest total transaction value. Whether the transaction values align with each customer's financial profile to prevent possible *money-laundering* risk.

Q2. This behavior may indicate are **higher trust in manual transactions**, especially for large sums. And **Limited adoption or comfort with digital banking** within this segment. The dominance of savings accounts also shows that many customers focus on liquidity rather than long-term investment products.

Q3. The Yogyakarta branch exhibits a significant concentration of high-value transactions, which may indicate:

- A large population of **priority or high-net-worth customers** in the region.
- Strong local economic activity from businesses and professionals.

However, such a high transaction volume also **increases operational and fraud risk** if not supported by proper monitoring systems.
The gap between Yogyakarta and other provinces suggests a **concentration of wealth** that could influence the bank's risk and marketing strategies.
Q3. **Top-performing branches**, especially Yogyakarta, highlight areas of strong economic activity that demand enhanced risk oversight and customized relationship management.

Q4. The transaction distribution appears **normal and well-controlled**, with no visible spikes or outlier values.

The interquartile range (IQR) is moderately wide, reflecting **healthy variability** in customer transaction behavior within expected boundaries.

The median lies roughly in the middle of the range, showing a **balanced spread** between lower and higher transaction groups without dominance from any extreme cluster.

Q5. **Balanced High-Value Behavior Across Genders:**

Both male and female customers engage in high-value transactions at a similar level.

This indicates that the bank's priority customer base is well diversified in terms of gender, and that financial capacity or trust in high-value banking products is equally shared.

Q5. Consistent Transaction Patterns Across Age Groups:

The almost uniform highest transaction value across age segments suggests that large transactions are not limited to a particular age demographic.

- Customers aged 51–65 likely represent experienced professionals and entrepreneurs with established wealth.
- Customers aged 25–50 may consist of rising professionals or business owners who are beginning to accumulate significant assets.
- Even younger customers (<25) are showing potential as emerging affluent individuals, possibly driven by digital entrepreneurship or early investment activity.

Q5. Implication for Outlier Detection:

Since the upper limit (~IDR 500 million) is consistent across demographics, potential anomalies may not depend on gender or age but on transaction patterns, frequency, or channel behavior.

This insight helps narrow the focus of fraud monitoring toward behavioral anomalies rather than demographic bias.

Recommendation

- **Risk Monitoring:** Implement real-time monitoring for large transactions by these top customers to detect unusual patterns. Implement real-time transaction monitoring and anomaly detection for top branches such as Yogyakarta and Riau(or lowest branches). **Develop automated** *outlier detection* systems to flag transactions above a certain threshold (example > Q3 + 1.5 × IQR).**Review flagged transactions** for compliance and AML (Anti-Money Laundering) checks to ensure legitimacy.
- **Fraud Prevention:** Develop a simple *anomaly detection model* comparing each customer's transaction against their historical behavior. **Establish alert mechanisms** for sudden, high-value transactions outside normal customer behavior. **Correlate high-value** transactions with location, time, and channel to detect potential fraud patterns (e.g., sudden changes in province or device).
- **Personalized Service:** Assign dedicated *relationship managers* and offer premium products (investment or wealth management) to retain these high-value customers while ensuring proper oversight.
- **Digital Adoption Program:** Educate and encourage priority customers to use Internet and Mobile Banking for better efficiency and security.
- **Product Diversification:** Promote investment and wealth-management products tailored to high-value customers.
- **Channel Optimization:** Enhance the digital experience with stronger authentication, faster support, and personalized assistance powered by AI or chat-based services.
- **Regional Risk Audit**: Conduct regular compliance and anti-money-laundering (AML) audits for branches with high financial activity.
- **Branch-Level Personalization:** Introduce exclusive programs and personalized financial services for high-value customers in high-activity regions to improve satisfaction and retention.
- **Personalized Handling for High-Value Customers** : Identify legitimate high-value clients and offer dedicated relationship management or investment advisory. Use behavioral segmentation to create tailored financial products and exclusive privileges.
- **Operational Insight:** Regularly monitor the evolution of transaction distribution — a widening gap between Q3 and upper outliers may indicate rising wealth concentration or potential risk exposure in certain branches.
- **Data-Driven Risk Segmentation:** Develop a behavioral-based risk model instead of purely demographic-based segmentation. Identify anomalies through transaction frequency, timing, or unusual cross-channel patterns rather than customer profile alone.
- **Enhanced Fraud Monitoring:** Integrate demographic context into fraud detection systems — e.g., flag large or unusual transactions if they deviate from the customer's historical norm, regardless of gender or age. Apply extra verification for unusually high transactions executed through non-typical channels (e.g., ATM or Mobile Banking for large amounts).
- **Personalized Customer Relationship Strategy:** For older customers (51–65) → offer Wealth Management or Retirement Investment Plans. For younger high-value clients (<35) → design Digital Investment and Flexible Savings solutions that appeal to their lifestyle. Maintain

gender-neutral premium engagement programs since both groups contribute equally to total high-value transactions.

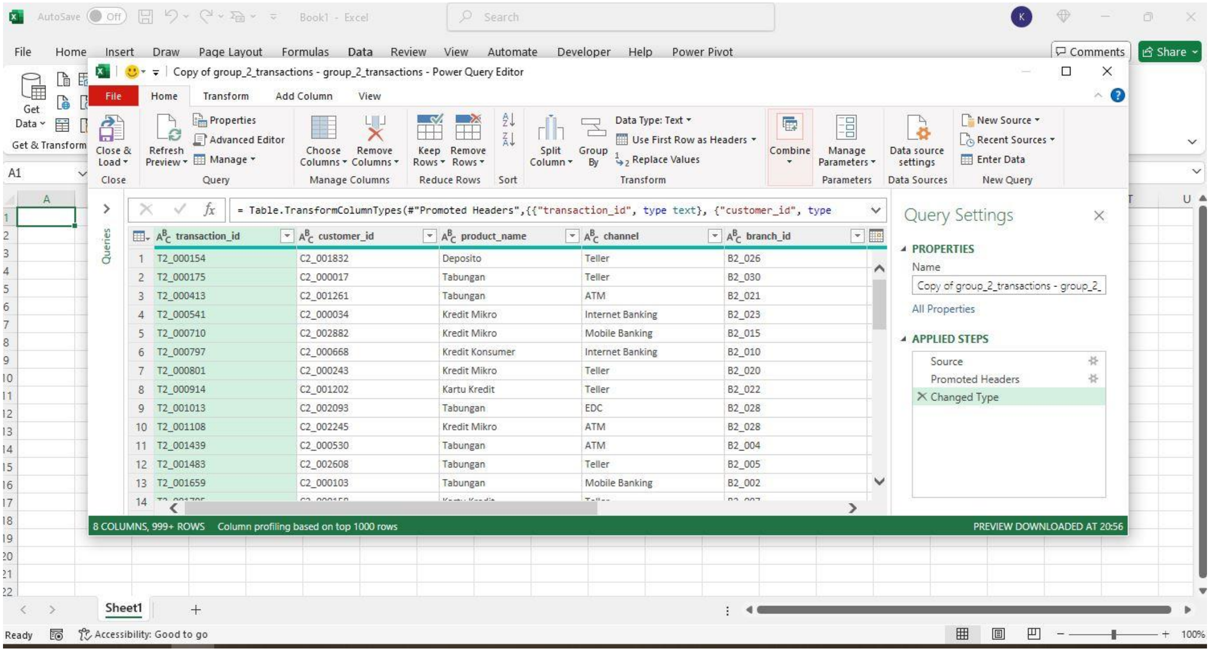
- **Operational Monitoring:** Continuously evaluate whether the transaction cap (~IDR 500 million) remains relevant for identifying outliers or needs adjustment as customer behavior evolves.

Section for Data Team

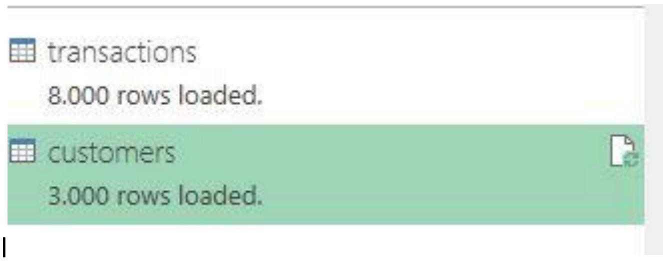
Merge Dataset with Excel

Merge 2 files using excel

1. Cleaning Transaction and Customer Data using Power Query



2. Merge 2 Data Mode



3. Merge Left Outer Transaction and Customer

Merge

Select tables and matching columns to create a merged table.

transactions

transaction_id	customer_id	product_name	channel	branch_id	transaction_date	amount	is_
T2_000154	C2_001832	Deposito	Teller	B2_026	03/05/2025	498517946	
T2_000175	C2_000017	Tabungan	Teller	B2_030	26/05/2024	497269804	
T2_000413	C2_001261	Tabungan	ATM	B2_021	16/05/2025	499990391	
T2_000541	C2_000034	Kredit Mikro	Internet Banking	B2_023	22/05/2023	496567056	

customers

customer_id	full_name	gender	age	province	segment	avg_balance	cc_limit
C2_000001	Mitra Najmudin	L	53	Bali	Priority	995397863	50000000
C2_000002	Panji Lailasari	L	52	Jawa Barat	Priority	688798071	100000000
C2_000003	Kariman Thamrin	P	41	Bali	Priority	674042970	20000000
C2_000004	Darmaji Dabukke	L	26	DKI Jakarta	Priority	771146801	50000000
C2_000005	Irnanto Namaga	P	30	Riau	Priority	785373995	50000000

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▷ Fuzzy matching options

✓ The selection matches 8000 of 8000 rows from the first table.

OK Cancel

4. Get a new dataset and upload it for use
https://raw.githubusercontent.com/iqbalyk/Miniproject-Digital-Skola/refs/heads/main/DatasetMiniproject3.csv

Analyzing with Python

Import Data

```
#import data csv TransactionCustomer
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

pd.set_option('display.max_columns', None)

df = pd.read_csv('https://raw.githubusercontent.com/iqbalyk/Miniproject-Digital-Skola/refs/heads/main/DatasetMiniproject3.csv', delimiter=';', engine='python')

# Change data type
cols_date = ['transaction_date']
for col in cols_date:
    df[col] = pd.to_datetime(df[col])

display(df.info())
df.head()
```

Import data and display to make it easier to see the displayed dataset.

Function

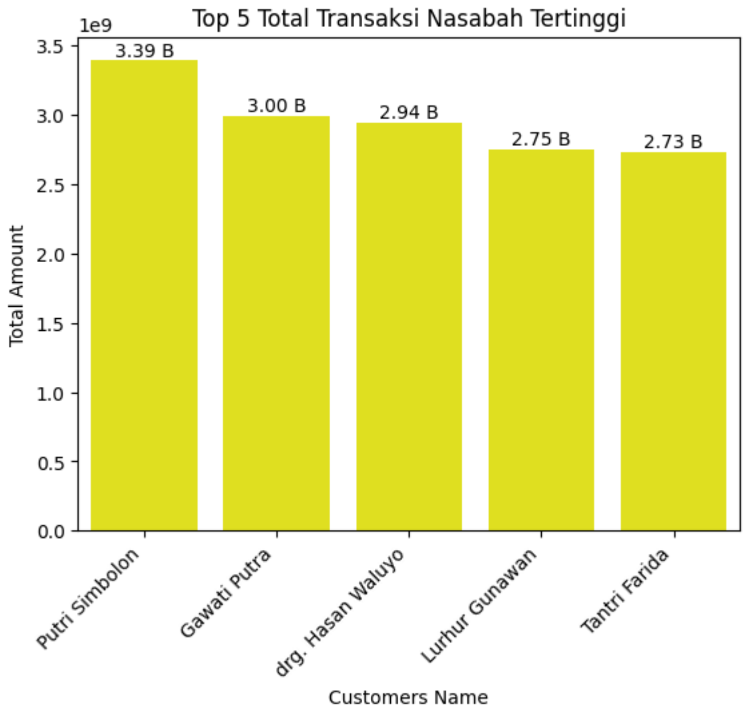
```
def simple_pivot_table(input_df, ind, val, agg):

  data = input_df.copy()

  pivot_a1 = pd.pivot_table(data, index=ind, values=val, aggfunc=agg).reset_index()
  pivot_a1 = pivot_a1.sort_values(by=val, ascending=False)
  return pivot_a1

def format_amount(amount):
    if amount >= 1_000_000_000:
        return f'{amount / 1_000_000_000:.2f} B'
    elif amount >= 1_000_000:
        return f'{amount / 1_000_000:.2f} M'
    else:
        return f'{amount:.2f}'

def bar_chart(input_df, x_axis, y_axis, xlabel, ylabel, title):

  ax= sns.barplot(data=input_df.head(5), x=x_axis, y=y_axis, color='yellow')
  plt.xlabel(xlabel)
  plt.xticks(rotation=45, ha='right')
  plt.ylabel(ylabel)
  plt.title(title)


  # Add Data Labels
  for container in ax.containers:
      ax.bar_label(container, fmt=lambda x: format_amount(x))

def bar_plot(input_df, x_axis, y_axis, hue, estimator, warna, legend, title):

    data4 = input_df.copy()

    ax2 = sns.barplot(data4, x=x_axis, y=y_axis, hue=hue,
                      estimator=estimator, palette=warna, legend=legend)
    plt.title(title)

    # Add Data Labels
    for container in ax2.containers:
        ax2.bar_label(container, fmt=lambda x: format_amount(x))
```

Code (Input & Output)

Q1. Who are the customers with the highest value transactions?

Answer:

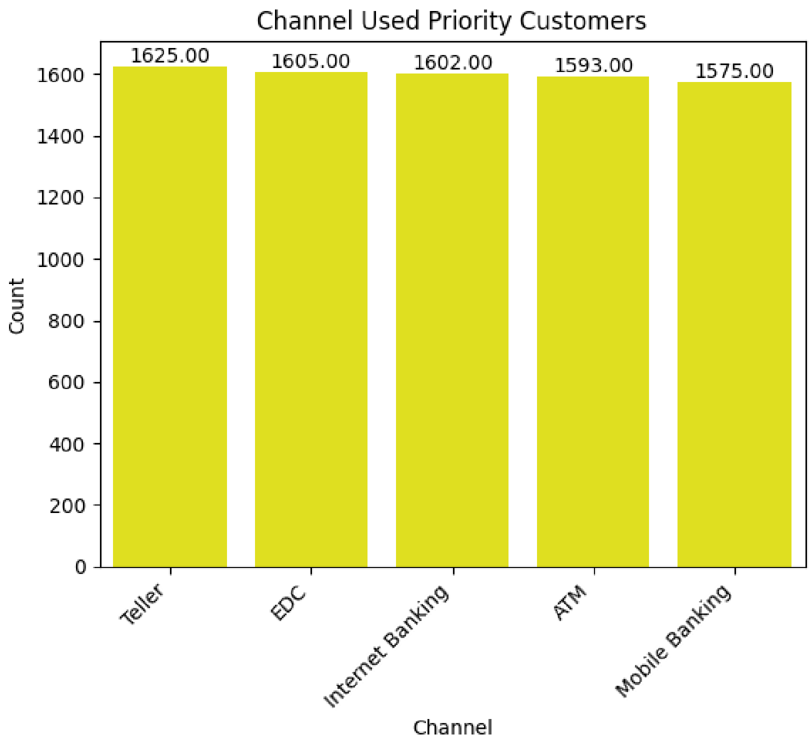
```
q1= df.copy()

pivot_a1 = simple_pivot_table(df, ind='full_name', val='amount', agg='sum')
bar_chart(pivot_a1, x_axis='full_name' , y_axis='amount' , title='Top 5 Total Transaksi Nasabah Tertinggi', xlabel= 'Customers Name', ylabel= 'Total Amount')
```



Q2. What products and channels do priority customers use most frequently?

Answer:

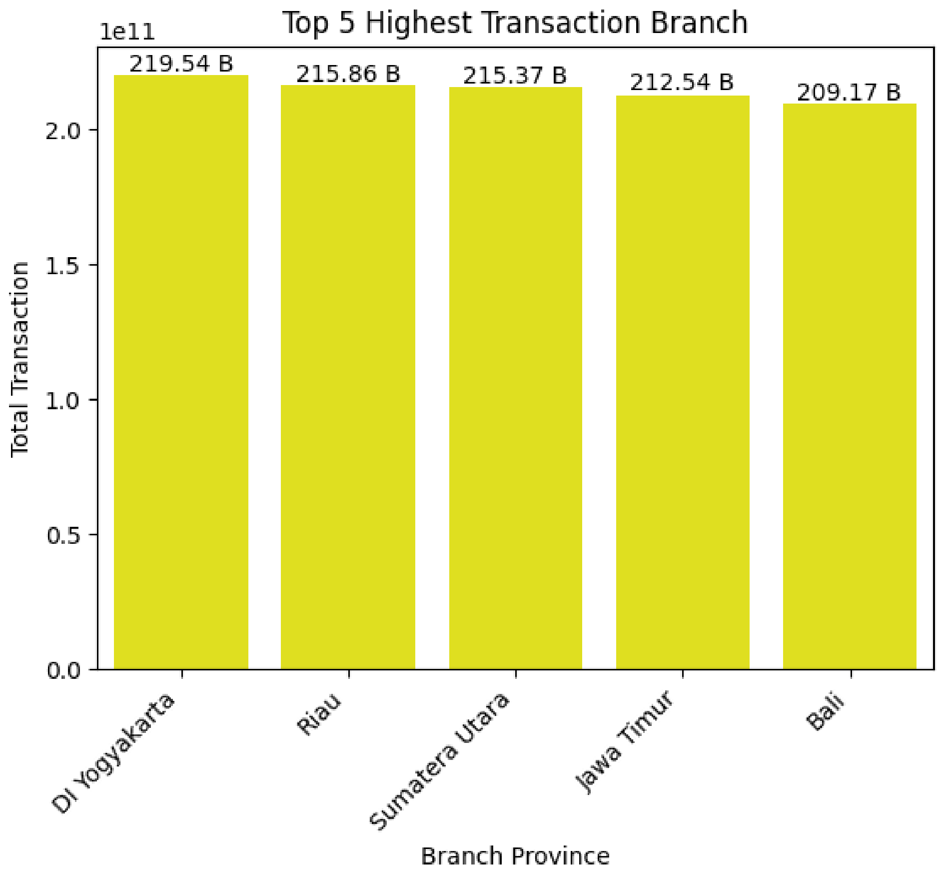
```
# Filter for priority customers
priority_customers_df = df[df['segment'] == 'Priority'].copy()

# Count the occurrences of each product
product_counts = priority_customers_df['product_name'].value_counts().reset_index(name='count')
product_counts = product_counts.rename(columns={'index': 'product_name'})

# Count the occurrences of each channel
channel_counts = priority_customers_df['channel'].value_counts().reset_index(name='count')
channel_counts = channel_counts.rename(columns={'index': 'channel'})

# Visualize product counts
bar_chart(product_counts, x_axis='product_name', y_axis='count', xlabel='Product Name', ylabel='Count', title='Product Used Priority Customers')
plt.show()

# Visualize channel counts
bar_chart(channel_counts, x_axis='channel', y_axis='count', xlabel='Channel', ylabel='Count', title='Channel Used Priority Customers')
plt.show()
```



Q3. Which branch did the highest value transactions occur?

Answer:

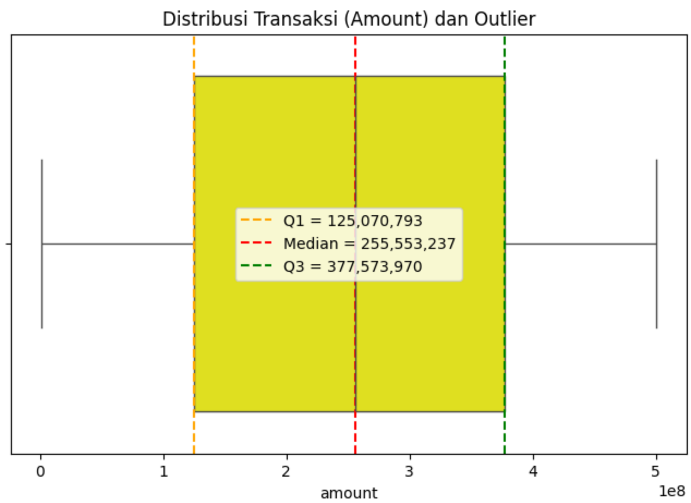
```
pivot_a1 = simple_pivot_table(df, ind='province', val='amount', agg='sum')
bar_chart(pivot_a1, x_axis='province' , y_axis='amount' , title='Top 5 Highest Transaction Branch', xlabel= 'Branch Province', ylabel= 'Total Transaction')
```



Q4. How to visualize transaction distribution and detect outliers with box plots?

Answer:

```
sns.boxplot(x=df["amount"], color="yellow")
plt.title("Distribusi Transaksi (Amount) dan Outlier")

plt.axvline(Q1, color='orange', linestyle='--', label=f"Q1 = {Q1:,.0f}")
plt.axvline(df["amount"].median(), color='red', linestyle='--', label=f"Median = {df['amount'].median():,.0f}")
plt.axvline(Q3, color='green', linestyle='--', label=f"Q3 = {Q3:,.0f}")
plt.legend()
plt.show()
```



Q4. What is the demographic profile of consumers who frequently make large transactions?

Answer:

```
# What is the demographic profile of consumers who frequently make large transactions?

df.groupby("gender", observed=False)["amount"].max().sort_values(ascending=True)
df.groupby("age_group", observed=False)["amount"].max().sort_values(ascending=True)

# Gender
bar_plot(df, x_axis='gender', y_axis='amount', hue='gender', estimator='max', warna='Set2', legend=False, title='Highest Transaction Per Gender')
plt.show()

# Age group
bar_plot(df, x_axis='age_group', y_axis='amount', hue='age_group', estimator='max', warna='Set2', legend=False, title='Highest Tracsaction By Group Age')
plt.show()
```